

Joint Control Barrier Function for Target Visibility and Thrust Deliverability

A camera-plane QP over the commanded inertial acceleration

Soft–Precise–Landing

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Scope. Visibility is fundamentally an *attitude* constraint (the body tilt shifts the field of view), but the attitude actually commanded is a byproduct of the desired inertial acceleration ${}^{\mathcal{I}}\mathbf{a}_d$ an outer control loop produces. Rather than constraining a separately-defined tilt variable and then separately deciding how much vertical thrust to spend, this note poses one projection directly over ${}^{\mathcal{I}}\mathbf{a}_d$ that jointly enforces (i) a field-of-view barrier condition and (ii) a thrust-deliverability bound, as a single constrained problem. The coupling between a commanded acceleration and the resulting feature motion is the **rotational part of the IBVS interaction matrix**, L_ω , evaluated at the measured image point — exact, *depth-independent*, no de-rotation or virtual frame required.

Notation

Symbol	Meaning
$\tilde{\mathbf{r}}_i, \tilde{r}_{i,k}$	measured camera-frame feature point i , tangent units, per axis k
$L_\omega(\tilde{\mathbf{r}}_i)$	rotational interaction matrix (roll/pitch columns), $\partial \dot{\tilde{\mathbf{r}}}_i / \partial \boldsymbol{\omega}_{\text{rp}}$
$\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the <i>rotation-axis</i> vector)
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	lean→rotation map: a lean in direction $\mathbf{y}$ rotates about $M\mathbf{y}$
$\tilde{L}_{\omega,i} = L_\omega M$	effective coupling from acceleration-ratio direction to feature velocity
${}^{\mathcal{I}}\mathbf{a}_d$	desired inertial acceleration; this projection’s decision variable
${}^{\mathcal{I}}a_{d,z}$	desired vertical acceleration component ($a_z \equiv {}^{\mathcal{I}}a_{d,z} $)
$\mathbf{y} = R_z(-\psi_b)({}^{\mathcal{I}}\mathbf{a}_{d,xy}/{}^{\mathcal{I}}a_{d,z})$	lateral-to-vertical acceleration ratio, expressed in image-axis coordinates
$\mathbf{y}_{\text{curr}}$	the same ratio, evaluated at the current realized attitude
A_{cap}	vehicle’s maximum achievable thrust-to-mass ratio
θ_{cap}	deliverable-tilt cap, $\ \mathbf{y}\ \leq \tan \theta_{\text{cap}}$
$\mathbf{d}_i$	exogenous (translation) feature drift, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\varphi_{\text{max}} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
τ, g	drift look-ahead; gravitational acceleration
κ_{az}	descent-rate relief gain (§3.1), acceleration per unit suppressed lean

1 Barrier on the real camera plane

Per image axis k , for feature point i ,

$$\boxed{h_{i,k} = \varphi_{\max,k} - |\tilde{\mathbf{r}}_{i,k}|}. \quad (1)$$

$h_{i,k} \geq 0 \iff$ feature point i is on the sensor on axis k . $\tilde{\mathbf{r}}_i$ is the *measured* camera-frame feature point — nothing is de-rotated.

2 Tilt→feature coupling, expressed in acceleration units

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point:

$$\dot{\tilde{\mathbf{r}}}_i = L_\omega(\tilde{\mathbf{r}}_i) \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}_i, \quad L_\omega = \left[\begin{array}{cc} xy & -(1+x^2) \\ 1+y^2 & -xy \end{array} \right]_{(x,y)=\tilde{\mathbf{r}}_i}. \quad (2)$$

L_ω contains **no depth**. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$ (using the true gyro rate), so it carries only genuine target motion, not our own tilting.

L_ω couples to a body *rotation rate* $\boldsymbol{\omega}_{\text{rp}}$, but the quantity a lateral-acceleration command directly determines is not a rotation rate — it is the underactuated quadrotor’s *lean*, the ratio of lateral to vertical thrust acceleration, ${}^{\mathcal{I}}\mathbf{a}_{\text{d},xy}/{}^{\mathcal{I}}a_{\text{d},z}$. Producing a steady lean in a given direction requires a rotation about the perpendicular axis; writing $\mathbf{y}$ for the acceleration ratio expressed in image-axis coordinates (a rotation by $R_z(-\psi_b)$ aligns the inertial x - y axes with the image axes),

$$\mathbf{y} = R_z(-\psi_b) \left(\frac{{}^{\mathcal{I}}\mathbf{a}_{\text{d},xy}}{{}^{\mathcal{I}}a_{\text{d},z}} \right), \quad \boldsymbol{\omega}_{\text{rp}} = M\mathbf{y}, \quad M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \quad (3)$$

$\mathbf{y}$ carries no information beyond the acceleration ratio itself — it is not a separate physical variable, only that ratio written in the coordinate frame L_ω is defined in. Substituting (3) into (2) gives the feature velocity directly as a function of the acceleration command:

$$\tilde{L}_{\omega,i}\mathbf{y} = \underbrace{\left(\frac{\tilde{L}_{\omega,i} R_z(-\psi_b)}{a_z} \right)}_{N_i(a_z)} {}^{\mathcal{I}}\mathbf{a}_{\text{d},xy}, \quad \tilde{L}_{\omega,i} = L_\omega M. \quad (4)$$

$N_i(a_z) = \tilde{L}_{\omega,i} R_z(-\psi_b)/a_z$ is the barrier’s Jacobian with respect to the lateral acceleration directly. It depends on a_z — the same quantity the joint projection below also solves for — which is exactly the coupling that projection is built to resolve, rather than treating a_z as an external given. For a finite acceleration change the exact map is the rotation homography of the measured point; the first-order form

$$\tilde{\mathbf{r}}_i({}^{\mathcal{I}}\mathbf{a}_{\text{d},xy}) \approx \tilde{\mathbf{r}}_i + N_i(a_z) {}^{\mathcal{I}}\mathbf{a}_{\text{d},xy} - \tilde{L}_{\omega,i} \mathbf{y}_{\text{curr}} \quad (5)$$

is the local linearization, accurate over one control step. $\mathbf{y}_{\text{curr}}$ is read directly from the realized attitude (not from ${}^{\mathcal{I}}\mathbf{a}_{\text{d}}$), so it needs no rotation from acceleration units.

3 The visibility–deliverability projection

Two physical constraints bind the same variable $\mathcal{I}\mathbf{a}_d$: the barrier condition of (1) along the feature response (2)–(4), and the requirement that the commanded thrust vector be one the vehicle can actually deliver. Writing $\mathcal{I}\mathbf{a}_d^*$ for the conditioned output,

$$\boxed{\begin{aligned} \mathcal{I}\mathbf{a}_d^* &= \arg \min_{\mathcal{I}\mathbf{a}_d} \|\mathcal{I}\mathbf{a}_d - \mathcal{I}\mathbf{a}_d^0\|^2 \\ \text{s.t. } |\tilde{\mathbf{r}}_i + N_i(a_z)\mathcal{I}\mathbf{a}_{d,xy} - \tilde{L}_{\omega,i}\mathbf{y}_{\text{curr}} + \tau\mathbf{d}_i| &\preceq \boldsymbol{\varphi}_{\text{max}}, \quad \|\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}} \end{aligned}} \quad (6)$$

where $\mathcal{I}\mathbf{a}_d^0$ is the unconstrained desired acceleration, the target this projection stays closest to. The first constraint is the per-feature FoV box, written directly on $\mathcal{I}\mathbf{a}_{d,xy}$; the second is the true thrust-deliverability sphere in gravity-shifted acceleration space (the vehicle’s total thrust vector is $m(\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3)$, and $\|\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}}$ is exactly the statement that this stays within the vehicle’s maximum). Both are convex for a fixed a_z ; since $N_i(a_z)$ itself depends on a_z , the two constraints are resolved jointly rather than sequentially: the lateral acceleration satisfying the box and the vertical acceleration satisfying the sphere are solved for together, not one after the other. $\mathcal{I}\mathbf{a}_d^*$ then defines the desired attitude and thrust magnitude for the inner attitude-tracking loop.

- **Directional** — only the outward-binding box row is active; inward (recovery) and tangential accelerations are free. No strangling.
- **Envelope-aware** — the deliverability side of (6) is the vehicle’s actual maximum thrust-to-mass ratio A_{cap} , not a fixed angular cap applied *as if* a_z were fixed at hover; it tightens or loosens correctly as a_z moves away from hover.
- **Forward-invariant** — the look-ahead shifts the box anchor by $\tau\mathbf{d}_i$, so the projection accounts for centroid drift over the next τ s.
- **Bounded by construction** — because $\mathcal{I}\mathbf{a}_d^*$ always satisfies the sphere constraint, the resulting acceleration ratio $\mathbf{y}^* = R_z(-\psi_b)\mathcal{I}\mathbf{a}_{d,xy}^*/a_z^*$ satisfies $\|\mathbf{y}^*\| \leq \tan\theta_{\text{cap}}$ at the solved a_z^* automatically; no separate cap application is needed.
- **Descent-rate aware** (§3.1) — while the box actively suppresses lateral authority, the vertical component is relaxed toward hover, in proportion to that suppression and clamped so it can only *slow* a descent, never reverse it; the trade lives inside (6) so the output stays box- and sphere-consistent.

3.1 Descent-rate relief under an active box

When the FoV box is binding it removes lateral acceleration the outer loop asked for: the realized lean $\mathbf{y}^*$ falls short of the desired $\mathbf{y}^0 = R_z(-\psi_b)(\mathcal{I}\mathbf{a}_{d,xy}^0/\mathcal{I}a_{d,z}^0)$ by

$$\Delta_\theta = \|\mathbf{y}^0 - \mathbf{y}^*\|, \quad (7)$$

the magnitude of lean the box could not grant this step. Pressing on at the full commanded closing rate while $\Delta_\theta > 0$ spends the remaining descent budget without the lateral error being corrected — the marker leaves the FoV, or touchdown arrives, before convergence. The projection therefore

spends part of that budget on time instead: the vertical component is pulled toward the hover value in proportion to Δ_θ ,

$$\boxed{\mathcal{I}a_{d,z}^* = \min\left(\mathcal{I}a_{d,z}, \max\left(\mathcal{I}a_{d,z}^0 - \kappa_{az} \Delta_\theta, -g\right)\right) \quad \text{when } \mathcal{I}a_{d,z}^0 > -g} \quad (8)$$

and left unchanged otherwise. The interpretation is exact per axis of (6): an A_{cap} budget spent laterally is not available vertically and vice versa; (8) makes the vertical side of that trade explicit whenever the box has forced the lateral side. Two bounds make it safe as a one-way relief:

- the inner $\max(\cdot, -g)$ clamps the result at the hover acceleration $-g$, so this term alone can bring a commanded descent toward zero closing rate but *never* past it into a commanded climb;
- the outer $\min(\mathcal{I}a_{d,z}, \cdot)$ keeps any lift the deliverability sphere or the outer loop already commanded ($\mathcal{I}a_{d,z} < -g$): the relief only ever adds toward-hover, it never removes lift.

The relief is recomputed from the unconstrained $\mathcal{I}a_{d,z}^0$ each outer iterate, so it does not accumulate across iterates, and it re-enters $N_i(a_z)$ on the next iterate — the lateral box and the vertical relief converge together, not one after the other. $\kappa_{az} = 0$ disables it and recovers (6) unchanged.

4 Feasibility

The box constraint alone is **infeasible** iff $\varphi_{\max,k} < 0$ for some k (the marker subtends the full FoV). The sphere constraint is always feasible by construction (it is satisfied by $\mathcal{I}\mathbf{a}_d = \mathbf{0}$). Genuine joint infeasibility — the box demanding more lateral acceleration than A_{cap} can deliver at any achievable a_z — reduces to

$$\sigma_{\min}(\tilde{L}_{\omega,i}) (\tan \theta_{\text{cap}} - \|\mathbf{y}_{\text{curr}}\|) \geq \max_{i,k} (|\tilde{r}_{i,k}| - \varphi_{\max,k})_+ + \bar{d} \quad \forall i, \quad (9)$$

evaluated at the solved a_z^* rather than at a fixed hover-assumed θ_{cap} : joint feasibility holds whenever some achievable a_z satisfies (9), a weaker requirement than demanding it hold at the hover value of a_z alone.

5 Assumptions

1. **The coupling $\tilde{L}_{\omega,i} = L_\omega M$ and its sign convention are calibrated.** The fixed 90° map $\boldsymbol{\omega}_{\text{rp}} = M\mathbf{y}$ bridges the acceleration-ratio direction and the rotation-axis rate that L_ω couples to.
2. **The conditioned acceleration is realized:** the inner attitude loop tracks the attitude built from $\mathcal{I}\mathbf{a}_d^*$ on a timescale fast relative to the outer loop.
3. **The drift d_i is the translation part** of the observed feature velocity, $\dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$, entering as *measured* motion — no separate disturbance term is introduced.
4. **First-order L_ω over the cycle.** Exact is the rotation homography of $\tilde{\mathbf{r}}_i$; the linearization (5) is accurate over one control step.
5. **A_{cap} is the vehicle's true achievable thrust-to-mass ratio**, used directly rather than a fixed angular margin computed from it at hover.

6 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the projection (6) contain only the measured image point $\tilde{\mathbf{r}}_i$, $\varphi_{\max} = \mathcal{R}/(2f)$, the body rate, the commanded acceleration, and the vehicle's own A_{cap} and g . L_ω **is depth-independent by construction**, and A_{cap} , g are fixed vehicle/physical constants, not measurements of target depth or altitude — the whole formulation remains manifestly scale- and depth-free.